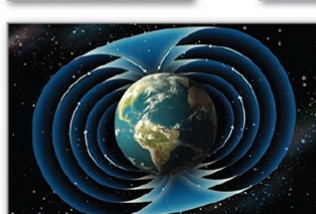


5 Inner Product Spaces



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1

5.1 Length and Dot Product in R^n

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Objectives

- Find the length of a vector and find a unit vector.
- Find the distance between two vectors.
- Find a dot product and the angle between two vectors, determine orthogonality, and verify the Cauchy-Schwarz Inequality, the triangle inequality, and the Pythagorean Theorem.
- Use a matrix product to represent a dot product.

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Vector Length and Unit Vectors

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Vector Length and Unit Vectors (1 of 3)

Definition of the Length of a Vector in R^n

The **length**, or **norm**, of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ in R^n is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

The length of a vector is also called its **magnitude**.

If $\|\mathbf{v}\| = 1$, then the vector $\mathbf{v}$ is a **unit vector**.

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Example 1 – The Length of a Vector in R^n (1 of 2)

a. In R^5 , the length of $\mathbf{v} = (0, -2, 1, 4, -2)$ is

$$\|\mathbf{v}\| = \sqrt{0^2 + (-2)^2 + 1^2 + 4^2 + (-2)^2} = \sqrt{25} = 5.$$

b. In R^3 , the length of $\mathbf{v} = (2/\sqrt{17}, -2/\sqrt{17}, 3/\sqrt{17})$ is

$$\begin{aligned} \|\mathbf{v}\| &= \sqrt{(2/\sqrt{17})^2 + (-2/\sqrt{17})^2 + (3/\sqrt{17})^2} \\ &= \sqrt{17/17} = 1. \end{aligned}$$

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Example 1 – The Length of a Vector in R^n (2 of 2)

The length of $\mathbf{v}$ is 1, so $\mathbf{v}$ is a unit vector, as shown in Figure 5.2.

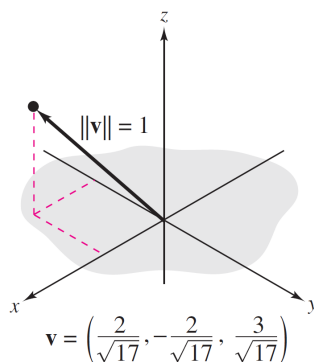


Figure 5.2

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Vector Length and Unit Vectors (2 of 3)

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are **parallel** when one is a scalar multiple of the other—that is, $\mathbf{u} = c\mathbf{v}$.

Moreover, if $c > 0$, then $\mathbf{u}$ and $\mathbf{v}$ have the **same direction**, and if $c < 0$, then $\mathbf{u}$ and $\mathbf{v}$ have **opposite directions**.

Theorem 5.1 Length of a scalar Multiple

Let $\mathbf{v}$ be a vector in R^n and let c be a scalar. Then

$$\|c\mathbf{v}\| = |c| \|\mathbf{v}\|$$

where $|c|$ is the absolute value of c .

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Vector Length and Unit Vectors (3 of 3)

Theorem 5.2 Unit Vector in the Direction of $\mathbf{v}$

If $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^n$, then the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

has length 1 and has the same direction as $\mathbf{v}$. This vector $\mathbf{u}$ is the **unit vector in the direction of $\mathbf{v}$** .

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Example 2 – Finding a Unit Vector (1 of 1)

Find the unit vector in the direction of $\mathbf{v} = (3, -1, 2)$, and verify that this vector has length 1.

Solution:

The unit vector in the direction of $\mathbf{v}$ is

$$\begin{aligned} \frac{\mathbf{v}}{\|\mathbf{v}\|} &= \frac{(3, -1, 2)}{\sqrt{3^2 + (-1)^2 + 2^2}} \\ &= \frac{1}{\sqrt{14}}(3, -1, 2) \\ &= \left(\frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right) \end{aligned}$$

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Example 2 – Solution (1 of 1)

which is a unit vector because

$$\sqrt{\left(\frac{3}{\sqrt{14}}\right)^2 + \left(-\frac{1}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2}$$

$$= \sqrt{\frac{14}{14}} = 1.$$

(See Figure 5.3.)

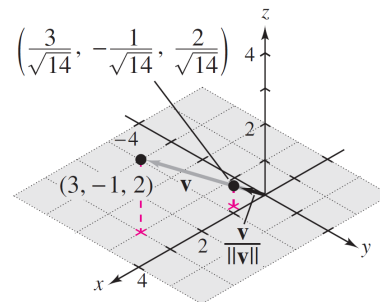


Figure 5.3

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Distance Between Two Vectors in $\mathbb{R}^n$

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Distance Between two Vectors in R^n (1 of 1)

Definition of Distance Between two Vectors

The distance between two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n is

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

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Example 3 – Finding the Distance Between two Vectors (1 of 2)

a. The distance between $\mathbf{u} = (-1, -4)$ and $\mathbf{v} = (2, 3)$ is

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \|(-1 - 2, -4 - 3)\| \\ &= \sqrt{(-3)^2 + (-7)^2} = \sqrt{58}. \end{aligned}$$

b. The distance between $\mathbf{u} = (0, 2, 2)$ and $\mathbf{v} = (2, 0, 1)$ is

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \|(0 - 2, 2 - 0, 2 - 1)\| \\ &= \sqrt{(-2)^2 + 2^2 + 1^2} = 3. \end{aligned}$$

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Example 3 – Finding the Distance Between two Vectors (2 of 2)

c. The distance between $\mathbf{u} = (3, -1, 0, -3)$ and $\mathbf{v} = (4, 0, 1, 2)$ is

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| \\ &= \|(3 - 4, -1 - 0, 0 - 1, -3 - 2)\| \\ &= \sqrt{(-1)^2 + (-1)^2 + (-1)^2 + (-5)^2} \\ &= \sqrt{28} \\ &= 2\sqrt{7}. \end{aligned}$$

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Dot Product and the Angle Between Two Vectors

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Dot Product and the Angle Between Two Vectors (1 of 7)

Definition of Dot Product in $\mathbb{R}^n$

The **dot product** of $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is the *scalar* quantity

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n.$$

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Dot Product and the Angle Between Two Vectors (2 of 7)

Theorem 5.3 Properties of the Dot Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$ and c is a scalar, then the properties listed below are true.

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
3. $c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v})$
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
5. $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

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Example 5 – Finding Dot Products (1 of 2)

Let $\mathbf{u} = (2, -2)$, $\mathbf{v} = (5, 8)$, and $\mathbf{w} = (-4, 3)$. Find each quantity.

- a. $\mathbf{u} \cdot \mathbf{v}$
- b. $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$
- c. $\mathbf{u} \cdot (2\mathbf{v})$
- d. $\|\mathbf{w}\|^2$
- e. $\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w})$

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Example 5 – Finding Dot Products (2 of 2)

Solution:

- a. By definition, you have

$$\mathbf{u} \cdot \mathbf{v} = 2(5) + (-2)(8) = -6.$$

- b. Using the result in part (a), you have

$$(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = -6\mathbf{w} = -6(-4, 3) = (24, -18).$$

- c. By Property 3 of Theorem 5.3, you have

$$\mathbf{u} \cdot (2\mathbf{v}) = 2(\mathbf{u} \cdot \mathbf{v}) = 2(-6) = -12.$$

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Example 5 – Solution (1 of 1)

d. By Property 4 of Theorem 5.3, you have

$$\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} = (-4)(-4) + (3)(3) = 25.$$

e. $2\mathbf{w} = (-8, 6)$, so you have

$$\mathbf{v} - 2\mathbf{w} = (5 - (-8), 8 - 6) = (13, 2).$$

Consequently,

$$\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w}) = 2(13) + (-2)(2) = 26 - 4 = 22.$$

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Example 6 – Using Properties of the Dot Product

Consider two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n such that $\mathbf{u} \cdot \mathbf{u} = 39$, $\mathbf{u} \cdot \mathbf{v} = -3$, and $\mathbf{v} \cdot \mathbf{v} = 79$.

Evaluate $(\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v})$.

Solution:

Using Theorem 5.3, rewrite the dot product as

$$\begin{aligned} (\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + \mathbf{v}) + (2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v}) \\ &= \mathbf{u} \cdot (3\mathbf{u}) + \mathbf{u} \cdot \mathbf{v} + (2\mathbf{v}) \cdot (3\mathbf{u}) + (2\mathbf{v}) \cdot \mathbf{v} \end{aligned}$$

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Example 6 – Solution (1 of 1)

$$\begin{aligned}
 &= 3(\mathbf{u} \cdot \mathbf{u}) + \mathbf{u} \cdot \mathbf{v} + 6(\mathbf{v} \cdot \mathbf{u}) + 2(\mathbf{v} \cdot \mathbf{v}) \\
 &= 3(\mathbf{u} \cdot \mathbf{u}) + 7(\mathbf{u} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{v}) \\
 &= 3(39) + 7(-3) + 2(79) \\
 &= 254.
 \end{aligned}$$

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Dot Product and the Angle Between Two Vectors (3 of 7)

Theorem 5.4 The Cauchy-Schwarz Inequality

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^n$, then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

where $|\mathbf{u} \cdot \mathbf{v}|$ denotes the *absolute value* of $\mathbf{u} \cdot \mathbf{v}$.

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Example 7 – Verifying the Cauchy-Schwarz Inequality (1 of 1)

Verify the Cauchy-Schwarz Inequality for $\mathbf{u} = (1, -1, 3)$ and $\mathbf{v} = (2, 0, -1)$.

Solution:

$\mathbf{u} \cdot \mathbf{v} = -1$, $\mathbf{u} \cdot \mathbf{u} = 11$, and $\mathbf{v} \cdot \mathbf{v} = 5$, so you have

$$|\mathbf{u} \cdot \mathbf{v}| = |-1| = 1$$

and

$$\begin{aligned}\|\mathbf{u}\| \|\mathbf{v}\| &= \sqrt{\mathbf{u} \cdot \mathbf{u}} \sqrt{\mathbf{v} \cdot \mathbf{v}} \\ &= \sqrt{11} \sqrt{5} \\ &= \sqrt{55}.\end{aligned}$$

The inequality $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ holds, because $1 \leq \sqrt{55}$. 25

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Dot Product and the Angle Between Two Vectors (4 of 7)

Definition of the Angle Between Two Vectors in R^n

The **angle** θ between two nonzero vectors in R^n can be found using

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi.$$

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Example 8 – Finding the angle Between two Vectors (1 of 1)

The angle between $\mathbf{u} = (-4, 0, 2, -2)$ and $\mathbf{v} = (2, 0, -1, 1)$ is

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{-12}{\sqrt{24}\sqrt{6}} = -\frac{12}{\sqrt{144}} = -1.$$

Consequently, $\theta = \pi$. It makes sense that $\mathbf{u}$ and $\mathbf{v}$ should have opposite directions, because $\mathbf{u} = 2\mathbf{v}$.

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Dot Product and the Angle Between Two Vectors (5 of 7)

Definition of orthogonal Vectors

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are **orthogonal** when

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

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Example 9 – Orthogonal Vectors in R^n (1 of 1)

a. The vectors $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = (0, 1, 0)$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (1)(0) + (0)(1) + (0)(0) = 0.$$

b. The vectors $\mathbf{u} = (3, 2, -1, 4)$ and $\mathbf{v} = (1, -1, 1, 0)$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (3)(1) + (2)(-1) + (-1)(1) + (4)(0) = 0.$$

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Example 10 – Finding orthogonal Vectors (1 of 1)

Determine all vectors in R^2 that are orthogonal to $\mathbf{u} = (4, 2)$.

Solution:

Let $\mathbf{v} = (v_1, v_2)$ be orthogonal to $\mathbf{u}$.

Then

$$\mathbf{u} \cdot \mathbf{v} = (4, 2) \cdot (v_1, v_2) = 4v_1 + 2v_2 = 0$$

which implies that $2v_2 = -4v_1$ and $v_2 = -2v_1$.

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Example 10 – Solution (1 of 1)

So, every vector that is orthogonal to $(4, 2)$ is of the form

$$\mathbf{v} = (t, -2t) = t(1, -2)$$

where t is a real number. (See Figure 5.4.)

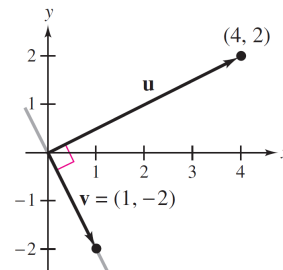


Figure 5.4

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Dot Product and the Angle Between Two Vectors (6 of 7)

The Cauchy-Schwarz Inequality can be used to prove another well-known inequality called the **triangle inequality**.

Theorem 5.5 The Triangle Inequality

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

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Dot Product and the Angle Between Two Vectors (7 of 7)

Theorem 5.6 The Pythagorean Theorem

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n , then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

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The Dot Product and Matrix Multiplication

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The Dot Product and Matrix Multiplication (1 of 1)

The dot product of two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

can be represented as the matrix product of the transpose of $\mathbf{u}$ multiplied by $\mathbf{v}$.

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = [u_1 v_1 + u_2 v_2 + \dots + u_n v_n]$$

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Example 11 – Using Matrix Multiplication to Find Dot Products (1 of 2)

a. The dot product of the vectors

$$\mathbf{u} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\text{is } \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = [(2)(3) + (0)(1)] = 6.$$

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Example 11 – Using Matrix Multiplication to Find Dot Products (2 of 2)

b. The dot product of the vectors

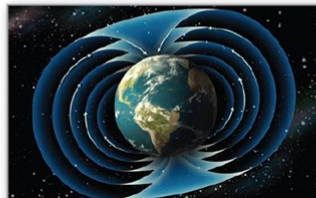
$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$$

$$\begin{aligned} \text{is } \mathbf{u} \cdot \mathbf{v} &= \mathbf{u}^T \mathbf{v} = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} \\ &= [(1)(3) + (2)(-2) + (-1)(4)] \\ &= -5. \end{aligned}$$

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5 Inner Product Spaces



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5.2 Inner Product Spaces

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Objectives

- Determine whether a function defines an inner product, and find the inner product of two vectors in R^n , $M_{m,n}$, P_n , and $C[a, b]$.
- Find an orthogonal projection of a vector onto another vector in an inner product space.

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Inner Products

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Inner Products (1 of 6)

The dot product, called the **Euclidean inner product**, is only one of several inner products that can be defined on R^n .

Definition of Inner Product

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in a vector space V , and let c be any scalar. An **inner product** on V is a function that associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ with each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ and satisfies the axioms listed below.

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$

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Inner Products (2 of 6)

Definition of Inner Product

3. $c\langle \mathbf{u}, \mathbf{v} \rangle = \langle c\mathbf{u}, \mathbf{v} \rangle$
4. $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

A vector space V with an inner product is an **inner product space**.

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Example 2 – A Different Inner Product for $\mathbb{R}^2$

Show that the function below defines an inner product on $\mathbb{R}^2$, where $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$.

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2$$

Solution:

1. The product of real numbers is commutative, so

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 2u_2 v_2 = v_1 u_1 + 2v_2 u_2 = \langle \mathbf{v}, \mathbf{u} \rangle.$$

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Example 2 – Solution (1 of 2)

2. Let $\mathbf{w} = (w_1, w_2)$ Then

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= u_1(v_1 + w_1) + 2u_2(v_2 + w_2) \\ &= u_1v_1 + u_1w_1 + 2u_2v_2 + 2u_2w_2 \\ &= (u_1v_1 + 2u_2v_2) + (u_1w_1 + 2u_2w_2) \\ &= \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle.\end{aligned}$$

3. If c is any scalar, then

$$c\langle \mathbf{u}, \mathbf{v} \rangle = c(u_1v_1 + 2u_2v_2) = (cu_1)v_1 + 2(cu_2)v_2 = \langle c\mathbf{u}, \mathbf{v} \rangle.$$

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Example 2 – Solution (2 of 2)

4. The square of a real number is nonnegative, so

$$\langle \mathbf{v}, \mathbf{v} \rangle = v_1^2 + 2v_2^2 \geq 0.$$

Moreover, this expression is equal to zero if and only if $\mathbf{v} = \mathbf{0}$ (that is, if and only if $v_1 = v_2 = 0$).

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Example 3 – A Function That is Not an Inner Product

Show that the function below is not an inner product on R^3 , where $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$.

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 - 2u_2 v_2 + u_3 v_3$$

Solution:

Observe that Axiom 4 theorem of 5.3 is not satisfied. For example, let $\mathbf{v} = (1, 2, 1)$.

Then $\langle \mathbf{v}, \mathbf{v} \rangle = (1)(1) - 2(2)(2) + (1)(1) = -6$, which is less than zero.

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Inner Products (3 of 6)

Theorem 5.7 Properties of Inner Products

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in an inner product space V , and let c be any real number.

1. $\langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. $\langle \mathbf{u}, c\mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$

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Inner Products (4 of 6)

Definitions of Length, Distance, and Angle

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V .

1. The **length** (or **norm**) of $\mathbf{u}$ is $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$.
2. The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
3. The **angle** between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ can be found using

$$\cos \theta = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}, \quad 0 \leq \theta \leq \pi.$$

4. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

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Inner Products (5 of 6)

If $\|\mathbf{u}\| = 1$, then $\mathbf{u}$ is a **unit vector**. Moreover, if $\mathbf{v}$ is any nonzero vector in an inner product space V , then the vector $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$ is the **unit vector in the direction of $\mathbf{v}$** .

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Example 6 – Finding Inner Products

For polynomials $p = a_0 + a_1x + \cdots + a_nx^n$ and $q = b_0 + b_1x + \cdots + b_nx^n$ in the vector space P_n , the function $\langle p, q \rangle = a_0b_0 + a_1b_1 + \cdots + a_nb_n$ is an inner product.

Let $p(x) = 1 - 2x^2$, $q(x) = 4 - 2x + x^2$, and $r(x) = x + 2x^2$ be polynomials in P_2 , and find each quantity.

- a. $\langle p, q \rangle$ b. $\langle q, r \rangle$ c. $\|q\|$ d. $d(p, q)$

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Example 6 – Solution (1 of 2)

- a. The inner product of p and q is

$$\langle p, q \rangle = a_0b_0 + a_1b_1 + a_2b_2 = (1)(4) + (0)(-2) + (-2)(1) = 2.$$

- b. The inner product of q and r is

$$\langle q, r \rangle = (4)(0) + (-2)(1) + (1)(2) = 0.$$

Notice that the vectors q and r are orthogonal.

- c. The length of q is $\|q\| = \sqrt{\langle q, q \rangle} = \sqrt{4^2 + (-2)^2 + 1^2} = \sqrt{21}.$

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Example 6 – Solution (2 of 2)

d. The distance between p and q is

$$\begin{aligned}
 d(p, q) &= \|p - q\| \\
 &= \|(1 - 2x^2) - (4 - 2x + x^2)\| \\
 &= \|-3 + 2x - 3x^2\| \\
 &= \sqrt{(-3)^2 + 2^2 + (-3)^2} \\
 &= \sqrt{22}.
 \end{aligned}$$

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Inner Products (6 of 6)

Theorem 5.8

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V .

1. Cauchy-Schwarz Inequality: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$
2. Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$
3. Pythagorean Theorem: $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

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Example 8 – An example of the Cauchy-Schwarz Inequality (Calculus)

Let $f(x) = 1$ and $g(x) = x$ be functions in the vector space $C[0, 1]$, with the inner product defined as:

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx$$

Verify that

$$|\langle f, g \rangle| \leq \|f\| \|g\|.$$

Solution:

For the left side of this inequality, you have

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx = \int_0^1 x dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}.$$

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Example 8 – Solution

For the right side of the inequality, you have

$$\|f\|^2 = \int_0^1 f(x)f(x) dx = \int_0^1 dx = \left. x \right|_0^1 = 1$$

and,

$$\|g\|^2 = \int_0^1 g(x)g(x) dx = \int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$

So,

$$\|f\| \|g\| = \sqrt{(1)\left(\frac{1}{3}\right)} = \frac{1}{\sqrt{3}} \approx 0.577, \text{ and } |\langle f, g \rangle| \leq \|f\| \|g\|.$$

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Orthogonal projections in inner product spaces

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Orthogonal projections in inner product spaces (1 of 2)

Definition of Orthogonal Projection

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V , such that $\mathbf{v} \neq \mathbf{0}$. Then the **orthogonal projection** of $\mathbf{u}$ onto $\mathbf{v}$ is.

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}.$$

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Example 10 – Finding an Orthogonal Projection in R^3

Use the Euclidean inner product in R^3 to find the orthogonal projection of $\mathbf{u} = (6, 2, 4)$ onto $\mathbf{v} = (1, 2, 0)$.

Solution:

$\mathbf{u} \cdot \mathbf{v} = 10$ and $\mathbf{v} \cdot \mathbf{v} = 5$, so the orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v} = \frac{10}{5} (1, 2, 0)$$

$$= 2(1, 2, 0) = (2, 4, 0)$$

as shown in Figure 5.9.

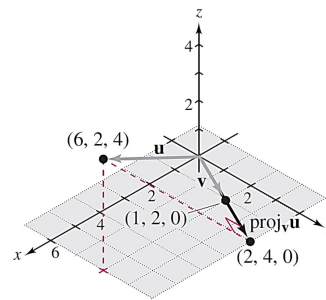


Figure 5.9

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Orthogonal projections in inner product spaces (2 of 2)

Theorem 5.9 Orthogonal Projection and Distance

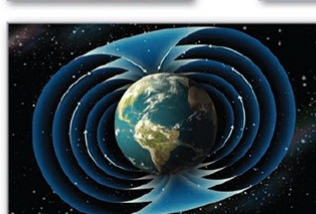
Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in an inner product space V , such that $\mathbf{v} \neq \mathbf{0}$. Then

$$d(\mathbf{u}, \text{proj}_{\mathbf{v}} \mathbf{u}) < d(\mathbf{u}, c\mathbf{v}), \quad c \neq \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

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5 Inner Product Spaces



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5.3

Orthonormal Bases: Gram-Schmidt Process

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Objectives

- Show that a set of vectors is orthogonal and forms an orthonormal basis, and represent a vector relative to an orthonormal basis.
- Apply the Gram-Schmidt orthonormalization process.

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Orthogonal and Orthonormal Sets

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Orthogonal and Orthonormal Sets (1 of 6)

Definitions of Orthogonal and Orthonormal Sets

A set S of vectors in an inner product space V is **orthogonal** when every pair of vectors in S is orthogonal. If, in addition, each vector in the set is a unit vector, then S is **orthonormal**.

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Orthogonal and Orthonormal Sets (2 of 6)

For $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, this definition has the form below.

- | <i>Orthogonal</i> | <i>Orthonormal</i> |
|---|---|
| 1. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0, i \neq j$ | 1. $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0, i \neq j$ |
| | 2. $\ \mathbf{v}_i\ = 1, i = 1, 2, \dots, n$ |

If S is a *basis*, then it is an **orthogonal basis** or an **orthonormal basis**, respectively.

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Example 1 – A nonstandard Orthonormal Basis for $\mathbb{R}^3$

Show that the set is an orthonormal basis for $\mathbb{R}^3$.

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right), \left(-\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3} \right), \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \right\}$$

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Example 1 – Solution (1 of 3)

First show that the three vectors are mutually orthogonal.

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -\frac{1}{6} + \frac{1}{6} + 0 = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = \frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} + 0 = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = -\frac{\sqrt{2}}{9} - \frac{\sqrt{2}}{9} + \frac{2\sqrt{2}}{9} = 0$$

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Example 1 – Solution (2 of 3)

Now, each vector is of length 1 because

$$\begin{aligned}\|v_1\| &= \sqrt{v_1 \cdot v_1} = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1 \\ \|v_2\| &= \sqrt{v_2 \cdot v_2} = \sqrt{\frac{1}{18} + \frac{1}{18} + \frac{8}{9}} = 1 \\ \|v_3\| &= \sqrt{v_3 \cdot v_3} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1.\end{aligned}$$

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Example 1 – Solution (3 of 3)

So, S is an orthonormal set. The three vectors do not lie in the same plane (see Figure 5.11), so you know that they span R^3 . By Theorem 4.12, they form a (nonstandard) orthonormal basis for R^3 .

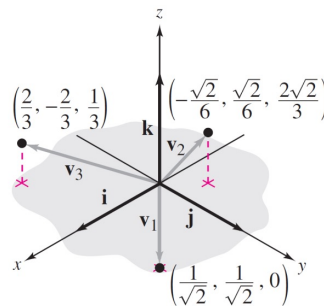


Figure 5.11

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Example 3 – An Orthogonal Set in $C[0, 2\pi]$ (Calculus)

In $C[0, 2\pi]$, with the inner product

$$\langle f, g \rangle = \int_0^{2\pi} f(x)g(x) dx$$

show that the set

$S = \{1, \sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin nx, \cos nx\}$ is orthogonal.

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Example 3 – Solution (1 of 2)

To show that this set is orthogonal, verify the inner products listed below, where m and n are positive integers.

$$\langle 1, \sin nx \rangle = \int_0^{2\pi} \sin nx dx = 0$$

$$\langle 1, \cos nx \rangle = \int_0^{2\pi} \cos nx dx = 0$$

$$\langle \sin mx, \cos nx \rangle = \int_0^{2\pi} \sin mx \cos nx dx = 0$$

$$\langle \sin mx, \sin nx \rangle = \int_0^{2\pi} \sin mx \sin nx dx = 0, \quad m \neq n$$

$$\langle \cos mx, \cos nx \rangle = \int_0^{2\pi} \cos mx \cos nx dx = 0, \quad m \neq n$$

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Example 3 – Solution (2 of 2)

One of these inner products is verified below. If $m \neq n$, then use the formula for rewriting a product of trigonometric functions as a sum to obtain

$$\int_0^{2\pi} \sin mx \cos nx \, dx = \frac{1}{2} \int_0^{2\pi} [\sin(m+n)x + \sin(m-n)x] \, dx = 0.$$

If $m = n$, then

$$\int_0^{2\pi} \sin nx \cos nx \, dx = \frac{1}{2n} \left[\sin^2 nx \right]_0^{2\pi} = 0.$$

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Orthogonal and Orthonormal Sets (3 of 6)

Theorem 5.10 Orthogonal Sets Are Linearly Independent

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthogonal set of *nonzero* vectors in an inner product space V , then S is linearly independent.

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Orthogonal and Orthonormal Sets (4 of 6)

Theorem 5.10 Corollary

If V is an inner product space of dimension n , then any orthogonal set of n nonzero vectors is a basis for V .

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Example 4 – Using Orthogonality to Test for a Basis

Show that the set S below is a basis for R^4 .

$$S = \{\overset{\mathbf{v}_1}{(2, 3, 2, -2)}, \overset{\mathbf{v}_2}{(1, 0, 0, 1)}, \overset{\mathbf{v}_3}{(-1, 0, 2, 1)}, \overset{\mathbf{v}_4}{(-1, 2, -1, 1)}\}$$

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Example 4 – Solution

The set S has four nonzero vectors. By the corollary to Theorem 5.10, you can show that S is a basis for R^4 by showing that it is an orthogonal set.

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 2 + 0 + 0 - 2 = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = -2 + 0 + 4 - 2 = 0$$

$$\mathbf{v}_1 \cdot \mathbf{v}_4 = -2 + 6 - 2 - 2 = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_3 = -1 + 0 + 0 + 1 = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_4 = -1 + 0 + 0 + 1 = 0$$

$$\mathbf{v}_3 \cdot \mathbf{v}_4 = 1 + 0 - 2 + 1 = 0$$

S is orthogonal, and by the corollary to Theorem 5.10, it is a basis for R^4 .

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Orthogonal and Orthonormal Sets (5 of 6)

Theorem 5.11 Coordinates Relative to an Orthonormal Basis

If $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V , then the coordinate representation of a vector $\mathbf{w}$ relative to B is

$$\mathbf{w} = \langle \mathbf{w}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{w}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \dots + \langle \mathbf{w}, \mathbf{v}_n \rangle \mathbf{v}_n.$$

In Theorem 5.11, the coordinates of $\mathbf{w}$ relative to the *orthonormal* basis B are called the **Fourier coefficients** of $\mathbf{w}$ relative to B , after Jean-Baptiste Joseph Fourier.

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Orthogonal and Orthonormal Sets (6 of 6)

The corresponding coordinate matrix of $\mathbf{w}$ relative to B is

$$[\mathbf{w}]_B = [c_1 \ c_2 \ \dots \ c_n]^T = [\langle \mathbf{w}, \mathbf{v}_1 \rangle \ \langle \mathbf{w}, \mathbf{v}_2 \rangle \ \dots \ \langle \mathbf{w}, \mathbf{v}_n \rangle]^T.$$

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Example 5 – Representing Vectors Relative to an Orthonormal Basis

Find the coordinate matrix of $\mathbf{w} = (5, -5, 2)$ relative to the orthonormal basis B for $\mathbb{R}^3$ below.

$$B = \left\{ \overset{\mathbf{v}_1}{\left(\frac{3}{5}, \frac{4}{5}, 0\right)}, \overset{\mathbf{v}_2}{\left(-\frac{4}{5}, \frac{3}{5}, 0\right)}, \overset{\mathbf{v}_3}{(0, 0, 1)} \right\}$$

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Example 5 – Solution

B is orthonormal (verify this), so use Theorem 5.11 to find the coordinates.

$$\mathbf{w} \cdot \mathbf{v}_1 = (5, -5, 2) \cdot \left(\frac{3}{5}, \frac{4}{5}, 0\right) = -1$$

$$\mathbf{w} \cdot \mathbf{v}_2 = (5, -5, 2) \cdot \left(-\frac{4}{5}, \frac{3}{5}, 0\right) = -7$$

$$\mathbf{w} \cdot \mathbf{v}_3 = (5, -5, 2) \cdot (0, 0, 1) = 2$$

So, the coordinate matrix relative to B is $[\mathbf{w}]_B = [-1 \ -7 \ 2]^T$.

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Gram-Schmidt Orthonormalization Process

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Gram-Schmidt Orthonormalization Process (1 of 7)

Having seen one of the advantages of orthonormal bases (the straightforwardness of coordinate representation), you will now look at a procedure for finding such a basis.

This procedure is called the **Gram-Schmidt orthonormalization process**, after the Danish mathematician Jorgen Pederson Gram (1850–1916) and the German mathematician Erhardt Schmidt (1876–1959).

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Gram-Schmidt Orthonormalization Process (2 of 7)

It has three steps.

1. Begin with a basis for the inner product space. It need not be orthogonal nor consist of unit vectors.
2. Convert the basis to an orthogonal basis.
3. Normalize each vector in the orthogonal basis to form an orthonormal basis.

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Gram-Schmidt Orthonormalization Process (3 of 7)

Theorem 5.12 Gram-Schmidt Orthonormalization Process

1. Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for an inner product space V .
2. Let $B' = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$, where

$$\mathbf{w}_1 = \mathbf{v}_1$$

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1$$

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Gram-Schmidt Orthonormalization Process (4 of 7)

Theorem 5.12 Gram-Schmidt Orthonormalization Process

$$\begin{aligned} \mathbf{w}_3 &= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 - \frac{\langle \mathbf{v}_3, \mathbf{w}_2 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2 \\ &\vdots \\ \mathbf{w}_n &= \mathbf{v}_n - \frac{\langle \mathbf{v}_n, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1 - \frac{\langle \mathbf{v}_n, \mathbf{w}_2 \rangle}{\langle \mathbf{w}_2, \mathbf{w}_2 \rangle} \mathbf{w}_2 - \dots - \frac{\langle \mathbf{v}_n, \mathbf{w}_{n-1} \rangle}{\langle \mathbf{w}_{n-1}, \mathbf{w}_{n-1} \rangle} \mathbf{w}_{n-1}. \end{aligned}$$

Then B' is an *orthogonal* basis for V .

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Gram-Schmidt Orthonormalization Process (5 of 7)

Theorem 5.12 Gram-Schmidt Orthonormalization Process

3. Let $\mathbf{u}_i = \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|}$. Then $B'' = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is an *orthonormal* basis for V .

Also, $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ for $k = 1, 2, \dots, n$.

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Example 6 – Applying the Gram-Schmidt Orthonormalization Process

Apply the Gram-Schmidt orthonormalization process to the basis B for $\mathbb{R}^2$ below.

$$B = \{\overset{\mathbf{v}_1}{(1, 1)}, \overset{\mathbf{v}_2}{(0, 1)}\}$$

Solution:

The Gram-Schmidt orthonormalization process produces

$$\mathbf{w}_1 = \mathbf{v}_1 = (1, 1)$$

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Example 6 – Solution (1 of 2)

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 = (0, 1) - \frac{1}{2}(1, 1) = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

The set $B' = \{\mathbf{w}_1, \mathbf{w}_2\}$ is an orthogonal basis for R^2 . By normalizing each vector in B' , you obtain

$$\mathbf{u}_1 = \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|} = \frac{1}{\sqrt{2}}(1, 1) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

$$\mathbf{u}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} = \frac{1}{1/\sqrt{2}}\left(-\frac{1}{2}, \frac{1}{2}\right) = \sqrt{2}\left(-\frac{1}{2}, \frac{1}{2}\right) = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

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Example 6 – Solution (2 of 2)

So, $B'' = \{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthonormal basis for R^2 . See Figure 5.14.

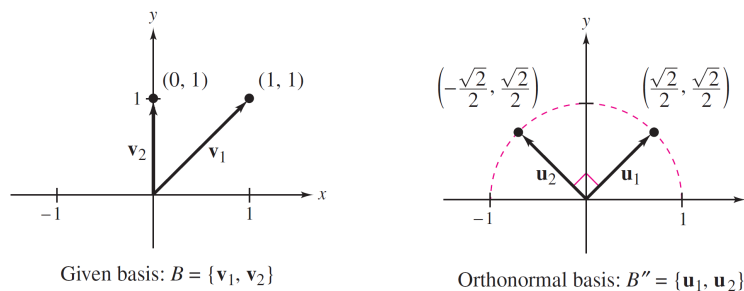


Figure 5.14

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Gram-Schmidt Orthonormalization Process (6 of 7)

The computations in the Gram-Schmidt orthonormalization process are sometimes simpler when you normalize each vector $\mathbf{w}_i$ *before* you use it to determine the next vector.

This **alternative form of the Gram-Schmidt orthonormalization process** has the steps listed below.

$$\mathbf{u}_1 = \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|} = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$$

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Gram-Schmidt Orthonormalization Process (7 of 7)

$$\mathbf{u}_2 = \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|}, \text{ where } \mathbf{w}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1$$

$$\mathbf{u}_3 = \frac{\mathbf{w}_3}{\|\mathbf{w}_3\|}, \text{ where } \mathbf{w}_3 = \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2$$

$$\vdots$$

$$\mathbf{u}_n = \frac{\mathbf{w}_n}{\|\mathbf{w}_n\|}, \text{ where } \mathbf{w}_n = \mathbf{v}_n - \langle \mathbf{v}_n, \mathbf{u}_1 \rangle \mathbf{u}_1 - \cdots - \langle \mathbf{v}_n, \mathbf{u}_{n-1} \rangle \mathbf{u}_{n-1}$$

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Example 10 – Alternative Form of the Gram-Schmidt Orthonormalization Process

Find an orthonormal basis for the solution space of the homogeneous linear system.

$$x_1 + x_2 + 7x_4 = 0$$

$$2x_1 + x_2 + 2x_3 + 6x_4 = 0$$

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Example 10 – Solution (1 of 4)

The augmented matrix for this system reduces as shown below.

$$\begin{bmatrix} 1 & 1 & 0 & 7 & 0 \\ 2 & 1 & 2 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -2 & 8 & 0 \end{bmatrix}$$

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Example 10 – Solution (2 of 4)

If you let $x_3 = s$ and $x_4 = t$, then each solution of the system has the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s + t \\ 2s - 8t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -8 \\ 0 \\ 1 \end{bmatrix}.$$

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Example 10 – Solution (3 of 4)

So, one basis for the solution space is

$$B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-2, 2, 1, 0), (1, -8, 0, 1)\}.$$

To find an orthonormal basis $B' = \{\mathbf{u}_1, \mathbf{u}_2\}$, use the alternative form of the Gram-Schmidt orthonormalization process, as shown below.

$$\begin{aligned} \mathbf{u}_1 &= \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \\ &= \left(-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right) \end{aligned}$$

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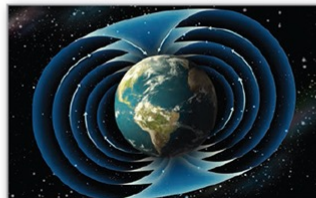
Example 10 – Solution (4 of 4)

$$\begin{aligned}
 \mathbf{w}_2 &= \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 \\
 &= (1, -8, 0, 1) - \left[(1, -8, 0, 1) \cdot \left(-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}, 0 \right) \right] \left(-\frac{2}{3}, \frac{2}{3}, \frac{1}{3}, 0 \right) \\
 &= (-3, -4, 2, 1) \\
 \mathbf{u}_2 &= \frac{\mathbf{w}_2}{\|\mathbf{w}_2\|} \\
 &= \left(-\frac{3}{\sqrt{30}}, -\frac{4}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}} \right)
 \end{aligned}$$

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5 Inner Product Spaces



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5.4**Mathematical Models and
Least Squares Analysis**

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Objectives

- Define the least squares problem.
- Find the orthogonal complement of a subspace and the projection of a vector onto a subspace.
- Find the four fundamental subspaces of a matrix.
- Solve a least squares problem.
- Use least squares for mathematical modeling.

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The Least Squares Problem

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Example 1 – Least Squares Regression Line (1 of 2)

Let $(1, 0)$, $(2, 1)$, and $(3, 3)$ be three points in R^2 . How can you find the line $y = c_0 + c_1x$ that “best fits” these points?

One way is to note that if the three points were collinear, then the system of equations below would be consistent.

$$C_0 + C_1 = 0$$

$$C_0 + 2C_1 = 1$$

$$C_0 + 3C_1 = 3$$

This system can be written in the matrix form $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}.$$

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Example 1 – Least Squares Regression Line (2 of 2)

The points are not collinear, however, so the system is inconsistent.

Although it is impossible to find $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$, you can look for an $\mathbf{x}$ that *minimizes* the norm of the error $\|A\mathbf{x} - \mathbf{b}\|$.

The solution $\mathbf{x} = [c_0 \ c_1]^T$ of this minimization problem results in the **least squares regression line** $y = c_0 + c_1x$.

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The Least Squares Problem (1 of 1)

Least Squares Problem

Given an $m \times n$ matrix A and a vector $\mathbf{b}$ in R^m , the **least squares problem** is to find $\mathbf{x}$ in R^n such that $\|A\mathbf{x} - \mathbf{b}\|^2$ is minimized.

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Orthogonal Subspaces

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Orthogonal Subspaces (1 of 6)

Definition of orthogonal subspaces

The subspaces S_1 and S_2 of R^n are **orthogonal** when $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ for all $\mathbf{v}_1$ in S_1 and all $\mathbf{v}_2$ in S_2 .

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Example 2 – Orthogonal Subspaces

The subspaces

$$S_1 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right\} \quad \text{and} \quad S_2 = \text{span}\left\{\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\right\}$$

are orthogonal because the dot product of any vector in S_1 and any vector in S_2 is zero.

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Orthogonal Subspaces (2 of 6)

Definition of Orthogonal Complement

If S is a subspace of R^n , then the **orthogonal complement** of S is the set

$$S^\perp = \{\mathbf{u} \in R^n : \mathbf{v} \cdot \mathbf{u} = 0 \text{ for all vectors } \mathbf{v} \in S\}.$$

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Example 3 – Finding the Orthogonal Complement

Find the orthogonal complement of the subspace S of R^4 spanned by the two column vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ of the matrix A .

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\mathbf{v}_1 \quad \mathbf{v}_2$

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Example 3 – Solution (1 of 2)

A vector $\mathbf{u} \in R^4$ is in the orthogonal complement of S when its dot product with each of the columns of A , $\mathbf{v}_1$ and $\mathbf{v}_2$, is zero. So, the orthogonal complement of S consists of all the vectors $\mathbf{u}$ such that $A^T \mathbf{u} = \mathbf{0}$.

$$A^T \mathbf{u} = \mathbf{0}$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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Example 3 – Solution (2 of 2)

That is, the orthogonal complement of S is the nullspace of the matrix A^T :

$$S^\perp = N(A^T).$$

Using the techniques for solving homogeneous linear systems, you can find that a basis for the orthogonal complement consists of the vectors

$$\mathbf{u}_1 = [-2 \quad 1 \quad 0 \quad 0]^T \quad \text{and} \quad \mathbf{u}_2 = [-1 \quad 0 \quad 1 \quad 0]^T.$$

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Orthogonal Subspaces (3 of 6)

Definition of Direct Sum

Let S_1 and S_2 be two subspaces of R^n . If each vector $\mathbf{x} \in R^n$ can be uniquely written as a sum of a vector $\mathbf{s}_1$ from S_1 and a vector $\mathbf{s}_2$ from S_2 , $\mathbf{x} = \mathbf{s}_1 + \mathbf{s}_2$, then R^n is the **direct sum** of S_1 and S_2 and you can write $R^n = S_1 \oplus S_2$.

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Example 4 – Direct Sum

a. From Example 2, R^3 is the direct sum of the subspaces

$$S_1 = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right\} \quad \text{and} \quad S_2 = \text{span}\left\{\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}\right\}.$$

b. From Example 3, you can see that $R^4 = S \oplus S^\perp$, where

$$S = \text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}\right\} \quad \text{and} \quad S^\perp = \text{span}\left\{\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}\right\}.$$

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Orthogonal Subspaces (4 of 6)

Theorem 5.13 Properties of Orthogonal Subspaces

Let S be a subspace of R^n . Then the properties listed below are true.

1. $\dim(S) + \dim(S^\perp) = n$
2. $R^n = S \oplus S^\perp$
3. $(S^\perp)^\perp = S$

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Orthogonal Subspaces (5 of 6)

Theorem 5.14 Projection onto a Subspace

If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_t\}$ is an orthonormal basis for the subspace S of R^n , and $\mathbf{v} \in R^n$, then

$$\text{proj}_S \mathbf{v} = (\mathbf{v} \cdot \mathbf{u}_1)\mathbf{u}_1 + (\mathbf{v} \cdot \mathbf{u}_2)\mathbf{u}_2 + \dots + (\mathbf{v} \cdot \mathbf{u}_t)\mathbf{u}_t.$$

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Example 5 – Projection Onto a Subspace

Find the projection of the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ onto the subspace S of R^3 spanned by the vectors

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{w}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

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Example 5 – Solution (1 of 2)

By normalizing $\mathbf{w}_1$ and $\mathbf{w}_2$ you obtain an orthonormal basis for S .

$$\{\mathbf{u}_1, \mathbf{u}_2\} = \left\{ \frac{1}{\sqrt{10}}\mathbf{w}_1, \frac{1}{2}\mathbf{w}_2 \right\} = \left\{ \begin{bmatrix} 0 \\ \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Use Theorem 5.14 to find the projection of $\mathbf{v}$ onto S .

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Example 5 – Solution (2 of 2)

$$\text{proj}_S \mathbf{v} = (\mathbf{v} \cdot \mathbf{u}_1)\mathbf{u}_1 + (\mathbf{v} \cdot \mathbf{u}_2)\mathbf{u}_2 = \frac{6}{\sqrt{10}} \begin{bmatrix} 0 \\ \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{9}{5} \\ \frac{3}{5} \end{bmatrix}$$

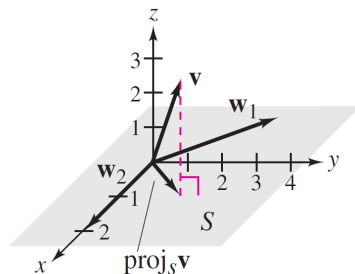


Figure 5.17

Figure 5.17 illustrates the projection of $\mathbf{v}$ onto the plane S .

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Orthogonal Subspaces (6 of 6)

Theorem 5.15 Orthogonal Projection and Distance

Let S be a subspace of R^n and let $\mathbf{v} \in R^n$. Then, for all $\mathbf{u} \in S$, $\mathbf{u} \neq \text{proj}_S \mathbf{v}$,

$$\|\mathbf{v} - \text{proj}_S \mathbf{v}\| < \|\mathbf{v} - \mathbf{u}\|.$$

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Fundamental Subspaces of a Matrix

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Fundamental Subspaces of a Matrix (1 of 2)

Recall that if A is an $m \times n$ matrix, then the column space of A is a subspace of $\mathbb{R}^m$ consisting of all vectors of the form $A\mathbf{x}$, $\mathbf{x} \in \mathbb{R}^n$. The four **fundamental subspaces** of the matrix A are listed below

$N(A)$ = nullspace of A $N(A^T)$ = nullspace of A^T

$R(A)$ = column space of A $R(A^T)$ = column space of A^T

These subspaces play a crucial role in the solution of the least squares problem.

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Example 6 – Fundamental Subspaces

Find the four fundamental subspaces of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solution:

The column space of A is simply the span of the first and third columns, because the second column is a scalar multiple of the first column.

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Example 6 – Solution (1 of 2)

The column space of A^T is equal to the row space of A , which is spanned by the first two rows. The nullspace of A is a solution space of the homogeneous system $A\mathbf{x} = \mathbf{0}$.

Finally, the nullspace of A^T is a solution space of the homogeneous system whose coefficient matrix is A^T .

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Example 6 – Solution (2 of 2)

A summary of these results is shown below.

$$\begin{aligned}
 R(A) &= \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\} & R(A^T) &= \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \\
 N(A) &= \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\} & N(A^T) &= \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}
 \end{aligned}$$

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Fundamental Subspaces of a Matrix (2 of 2)

Theorem 5.16 Fundamental Subspaces of a Matrix

If A is an $m \times n$ matrix, then

1. $R(A)$ and $N(A^T)$ are orthogonal subspaces of R^m .
2. $R(A^T)$ and $N(A)$ are orthogonal subspaces of R^n .
3. $R(A) \oplus N(A^T) = R^m$.
4. $R(A^T) \oplus N(A) = R^n$.

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Solving the Least Squares Problem

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Example 7 – Finding the Least Squares Solution

Find the solution of the least squares problem from

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$$

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Example 7 – Solution (1 of 2)

Begin by finding the matrix products below.

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$$

The normal equations are represented by the system

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 4 \\ 11 \end{bmatrix}.$$

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Example 7 – Solution (2 of 2)

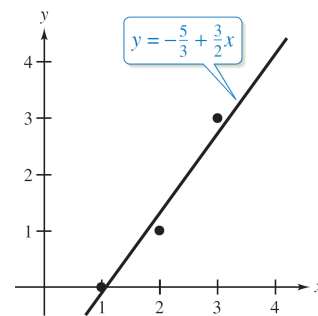
The solution of this system of equations is

$$\mathbf{x} = \begin{bmatrix} -\frac{5}{3} \\ \frac{3}{2} \end{bmatrix}$$

which implies that the least squares regression line for the data is

$$y = -\frac{5}{3} + \frac{3}{2}x,$$

as shown in the figure.



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Example 8 – Orthogonal Projection Onto a Subspace

Find the orthogonal projection of the vector

$$\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

onto the column space S of the matrix

$$A = \begin{bmatrix} 0 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}.$$

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Example 8 – Solution (1 of 3)

To find the orthogonal projection of $\mathbf{b}$ onto S , first solve the least squares problem $A\mathbf{x} = \mathbf{b}$.

Find the matrix products $A^T A$ and $A^T \mathbf{b}$.

$$\begin{aligned} A^T A &= \begin{bmatrix} 0 & 3 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 0 \\ 0 & 4 \end{bmatrix} \end{aligned}$$

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Example 8 – Solution (2 of 3)

$$\begin{aligned} A^T \mathbf{b} &= \begin{bmatrix} 0 & 3 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ 2 \end{bmatrix} \end{aligned}$$

The normal equations are represented by the system

$$\begin{aligned} A^T A \mathbf{x} &= A^T \mathbf{b} \\ \begin{bmatrix} 10 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 6 \\ 2 \end{bmatrix}. \end{aligned}$$

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Example 8 – Solution (3 of 3)

The solution of these equations is

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ \frac{1}{2} \end{bmatrix}.$$

Finally, the projection of $\mathbf{b}$ onto S is

$$A\mathbf{x} = \begin{bmatrix} 0 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{3}{5} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{9}{5} \\ \frac{3}{5} \end{bmatrix}$$

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Mathematical Modelling

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Mathematical Modelling (1 of 1)

Least squares problems play a fundamental role in mathematical modelling of real-life phenomena.

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Example 9 – World Population

The table shows the world population (in billions) for six different years. (Source: U.S. Census Bureau)

Year	1985	1990	1995	2000	2005	2010
Population, y	4.9	5.3	5.7	6.1	6.5	6.9

Let $x = 5$ represent the year 1985. Find the least squares regression quadratic polynomial $y = c_0 + c_1x + c_2x^2$ for the data and use the model to estimate the population for the year 2020.

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Example 9 – Solution (1 of 4)

By substituting the data points (5, 4.9), (10, 5.3), (15, 5.7), (20, 6.1), (25, 6.5), and (30, 6.9) into the quadratic polynomial $y = c_0 + c_1x + c_2x^2$, you obtain the system of linear equations below.

$$C_0 + 5C_1 + 25C_2 = 4.9$$

$$C_0 + 10C_1 + 100C_2 = 5.3$$

$$C_0 + 15C_1 + 225C_2 = 5.7$$

$$C_0 + 20C_1 + 400C_2 = 6.1$$

$$C_0 + 25C_1 + 625C_2 = 6.5$$

$$C_0 + 30C_1 + 900C_2 = 6.9$$

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Example 9 – Solution (2 of 4)

This produces the least squares problem

$$Ax = b$$

$$\begin{bmatrix} 1 & 5 & 25 \\ 1 & 10 & 100 \\ 1 & 15 & 225 \\ 1 & 20 & 400 \\ 1 & 25 & 625 \\ 1 & 30 & 900 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4.9 \\ 5.3 \\ 5.7 \\ 6.1 \\ 6.5 \\ 6.9 \end{bmatrix}$$

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Example 9 – Solution (3 of 4)

The normal equations are represented by the system

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

$$\begin{bmatrix} 6 & 105 & 2275 \\ 105 & 2275 & 55,125 \\ 2275 & 55,125 & 1,421,875 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 35.4 \\ 654.5 \\ 14,647.5 \end{bmatrix}$$

and their solution is $\mathbf{x} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 0.08 \\ 0 \end{bmatrix}$.

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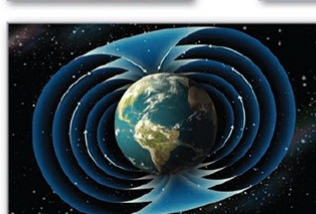
Example 9 – Solution (4 of 4)

Note that $c_2 = 0$. So, the least squares polynomial is the *linear* polynomial $y = 4.5 + 0.08x$. Evaluating this polynomial at $x = 40$ gives the estimate of the world population for the year 2020: $y = 4.5 + 0.08(40) = 7.7$ billion.

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5 Inner Product Spaces



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5.5 Applications of Inner Product Spaces

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Objectives

- Find the cross product of two vectors in $\mathbb{R}^3$.
- Find the linear or quadratic least squares approximation of a function.
- Find the n th-order Fourier approximation of a function.

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The cross product of two vectors
in $\mathbb{R}^3$

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The cross product of two vectors in R^3 (1 of 3)

Definition of the Cross Product of Two Vectors

Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ be vectors in R^3 . The **cross product** of $\mathbf{u}$ and $\mathbf{v}$ is the vector

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k}.$$

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Example 1 – Finding the Cross Product of Two Vectors

Let $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. Find each cross product.

a. $\mathbf{u} \times \mathbf{v}$ **b. $\mathbf{v} \times \mathbf{u}$** **c. $\mathbf{v} \times \mathbf{v}$**

Solution

$$\begin{aligned} \text{a. } \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 3 & 1 & -2 \end{vmatrix} \\ &= \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} \mathbf{k} \\ &= 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k} \end{aligned}$$

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Example 1 – Solution (1 of 2)

$$\begin{aligned}
 \text{b. } \mathbf{v} \times \mathbf{u} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -2 \\ 1 & -2 & 1 \end{vmatrix} \\
 &= \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{k} \\
 &= -3\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}
 \end{aligned}$$

Note that this result is the negative of that in part (a).

$$\text{c. } \mathbf{v} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -2 \\ 3 & 1 & -2 \end{vmatrix}$$

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Example 1 – Solution (2 of 2)

$$\begin{aligned}
 &= \begin{vmatrix} 1 & -2 \\ 1 & -2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & -2 \\ 3 & -2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} \mathbf{k} \\
 &= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}
 \end{aligned}$$

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The cross product of two vectors in R^3 (2 of 3)

Theorem 5.17 Algebraic Properties of the Cross Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in R^3 and c is a scalar, then the properties listed below are true.

1. $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$
3. $c(\mathbf{u} \times \mathbf{v}) = c\mathbf{u} \times \mathbf{v} = \mathbf{u} \times c\mathbf{v}$
4. $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$
5. $\mathbf{u} \times \mathbf{u} = \mathbf{0}$
6. $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$

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The cross product of two vectors in R^3 (3 of 3)

Theorem 5.18 Geometric Properties of the Cross Product

If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in R^3 , then the properties listed below are true.

1. $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.
2. The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is found using

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$
3. $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.
4. The parallelogram having $\mathbf{u}$ and $\mathbf{v}$ as adjacent sides has an area of $\|\mathbf{u} \times \mathbf{v}\|$.

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Example 2 – Finding a Vector orthogonal to two given Vectors

Find a unit vector orthogonal to both

$$\mathbf{u} = \mathbf{i} - 4\mathbf{j} + \mathbf{k}$$

and

$$\mathbf{v} = 2\mathbf{i} + 3\mathbf{j}.$$

Solution

From Property 1 of Theorem 5.18, you know that the cross product

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -4 & 1 \\ 2 & 3 & 0 \end{vmatrix} = -3\mathbf{i} + 2\mathbf{j} + 11\mathbf{k}$$

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Example 2 – Solution (1 of 3)

Is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, as shown in Figure 5.21.

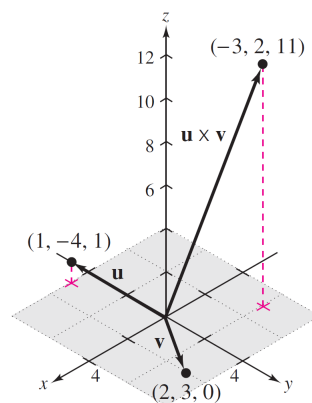


Figure 5.21

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Example 2 – Solution (2 of 3)

Then, by dividing by the length of $\mathbf{u} \times \mathbf{v}$,

$$\|\mathbf{u} \times \mathbf{v}\| = \sqrt{(-3)^2 + 2^2 + 11^2} = \sqrt{134}$$

you obtain the unit vector

$$\frac{\mathbf{u} \times \mathbf{v}}{\|\mathbf{u} \times \mathbf{v}\|} = -\frac{3}{\sqrt{134}}\mathbf{i} + \frac{2}{\sqrt{134}}\mathbf{j} + \frac{11}{\sqrt{134}}\mathbf{k}$$

which is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, because

$$\left(-\frac{3}{\sqrt{134}}, \frac{2}{\sqrt{134}}, \frac{11}{\sqrt{134}}\right) \cdot (1, -4, 1) = 0$$

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Example 2 – Solution (3 of 3)

And,

$$\left(-\frac{3}{\sqrt{134}}, \frac{2}{\sqrt{134}}, \frac{11}{\sqrt{134}}\right) \cdot (2, 3, 0) = 0.$$

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Least Squares Approximations (Calculus)

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Least Squares Approximations (Calculus) (1 of 2)

Definition of Least Squares Approximation

Let f be continuous on $[a, b]$, and let W be a subspace of $C[a, b]$.

A function g in W is the **least squares approximation** of f with respect to W when the value of

$$I = \int_a^b [f(x) - g(x)]^2 dx$$

is a minimum with respect to all other functions in W .

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Example 4 – Finding a Least Squares Approximation

Find the least squares approximation $g(x) = a_0 + a_1x$ of

$$f(x) = e^x, \quad 0 \leq x \leq 1.$$

Solution

For this approximation, you need to find the constants a_0 and a_1 that minimize the value of

$$\begin{aligned} I &= \int_0^1 [f(x) - g(x)]^2 dx \\ &= \int_0^1 (e^x - a_0 - a_1x)^2 dx. \end{aligned}$$

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Example 4 – Solution (1 of 3)

Evaluating this integral, you have

$$\begin{aligned} I &= \int_0^1 (e^x - a_0 - a_1x)^2 dx \\ &= \int_0^1 (e^{2x} - 2a_0e^x - 2a_1xe^x + a_0^2 + 2a_0a_1x + a_1^2x^2) dx \\ &= \left[\frac{1}{2}e^{2x} - 2a_0e^x - 2a_1e^x(x-1) + a_0^2x + a_0a_1x^2 + a_1^2\frac{x^3}{3} \right]_0^1 \\ &= \frac{1}{2}(e^2 - 1) - 2a_0(e - 1) - 2a_1 + a_0^2 + a_0a_1 + \frac{1}{3}a_1^2. \end{aligned}$$

Now, considering I to be a function of the variables a_0 and a_1 , use calculus to determine the values of a_0 and a_1 that minimize I .

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Example 4 – Solution (2 of 3)

Specifically, by setting the partial derivatives

$$\frac{\partial I}{\partial a_0} = 2a_0 - 2e + 2 + a_1$$

$$\frac{\partial I}{\partial a_1} = a_0 + \frac{2}{3}a_1 - 2$$

equal to zero, you obtain the two linear equations in a_0 and a_1 below.

$$2a_0 + a_1 = 2(e - 1)$$

$$3a_0 + 2a_1 = 6$$

The solution of this system is

$$a_0 = 4e - 10 \approx 0.873 \quad \text{and} \quad a_1 = 18 - 6e \approx 1.690.$$

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Example 4 – Solution (3 of 3)

(Verify this.) So, the best *linear approximation* of $f(x) = e^x$ on the interval $[0, 1]$ is

$$g(x) = 4e - 10 + (18 - 6e)x \approx 0.873 + 1.690x.$$

Figure 5.23 shows the graphs of f and g on $[0, 1]$.

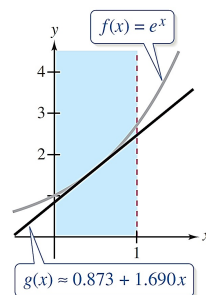


Figure 5.23

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Least Squares Approximations (Calculus) (2 of 2)

Theorem 5.19 Least Squares Approximation

Let f be continuous on $[a, b]$, and let W be a finite-dimensional subspace of $C[a, b]$.

The least squares approximating function of f with respect to W is

$$g = \langle f, \mathbf{w}_1 \rangle \mathbf{w}_1 + \langle f, \mathbf{w}_2 \rangle \mathbf{w}_2 + \cdots + \langle f, \mathbf{w}_n \rangle \mathbf{w}_n$$

where $B = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ is an orthonormal basis for W .

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Example 6 – Finding a Least Squares Approximation

Find the least squares approximation of $f(x) = \sin x$, $0 \leq x \leq \pi$, with respect to the subspace W of polynomial functions of degree 2 or less.

Solution

To use Theorem 5.19, apply the Gram-Schmidt orthonormalization process to the standard basis for

W , $\{1, x, x^2\}$, to obtain the orthonormal basis

$$B = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} = \left\{ \frac{1}{\sqrt{\pi}}, \frac{\sqrt{3}}{\pi\sqrt{\pi}}(2x - \pi), \frac{\sqrt{5}}{\pi^2\sqrt{\pi}}(6x^2 - 6\pi x + \pi^2) \right\}.$$

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Example 6 – Solution (1 of 2)

The least squares approximating function g is

$$g(x) = \langle f, \mathbf{w}_1 \rangle \mathbf{w}_1 + \langle f, \mathbf{w}_2 \rangle \mathbf{w}_2 + \langle f, \mathbf{w}_3 \rangle \mathbf{w}_3$$

and you have

$$\langle f, \mathbf{w}_1 \rangle = \frac{1}{\sqrt{\pi}} \int_0^{\pi} \sin x \, dx = \frac{2}{\sqrt{\pi}}$$

$$\langle f, \mathbf{w}_2 \rangle = \frac{\sqrt{3}}{\pi \sqrt{\pi}} \int_0^{\pi} \sin x (2x - \pi) \, dx = 0$$

$$\langle f, \mathbf{w}_3 \rangle = \frac{\sqrt{5}}{\pi^2 \sqrt{\pi}} \int_0^{\pi} \sin x (6x^2 - 6\pi x + \pi^2) \, dx = \frac{2\sqrt{5}}{\pi^2 \sqrt{\pi}} (\pi^2 - 12).$$

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Example 6 – Solution (2 of 2)

So, g is

$$g(x) = \frac{2}{\pi} + \frac{10(\pi^2 - 12)}{\pi^5} (6x^2 - 6\pi x + \pi^2) \approx -0.4177x^2 + 1.3122x - 0.0505.$$

Figure 5.25 shows the graphs of f and g .

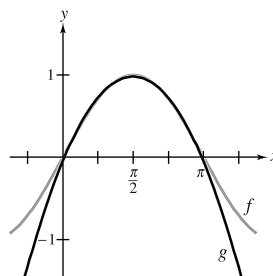


Figure 5.25

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Fourier Approximations (Calculus)

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Fourier Approximations (Calculus) (1 of 2)

Theorem 5.20 Fourier Approximation

On the interval $[0, 2\pi]$, the least squares approximation of a continuous function f with respect to the vector space spanned by

$$\{1, \cos x, \dots, \cos nx, \sin x, \dots, \sin nx\}$$

is

$$g(x) = \frac{a_0}{2} + a_1 \cos x + \dots + a_n \cos nx + b_1 \sin x + \dots + b_n \sin nx$$

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Fourier Approximations (Calculus) (2 of 2)

Theorem 5.20 Fourier Approximation

where the **Fourier coefficients** $a_0, a_1, \dots, a_n, b_1, \dots, b_n$ are

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_j = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos jx dx, \quad j = 1, 2, \dots, n$$

$$b_j = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin jx dx, \quad j = 1, 2, \dots, n.$$

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Example 7 – Finding a Fourier Approximation

Find the third-order Fourier approximation of $f(x) = x, 0 \leq x \leq 2\pi$.

Solution

Using Theorem 5.20, you have

$$g(x) = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x$$

where

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} x dx = \frac{1}{\pi} 2\pi^2 = 2\pi$$

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Example 7 – Solution (1 of 2)

$$a_j = \frac{1}{\pi} \int_0^{2\pi} x \cos jx \, dx = \left[\frac{1}{\pi j^2} \cos jx + \frac{x}{\pi j} \sin jx \right]_0^{2\pi} = 0$$

$$b_j = \frac{1}{\pi} \int_0^{2\pi} x \sin jx \, dx = \left[\frac{1}{\pi j^2} \sin jx - \frac{x}{\pi j} \cos jx \right]_0^{2\pi} = -\frac{2}{j}.$$

This implies that $a_0 = 2\pi$, $a_1 = 0$, $a_2 = 0$, $a_3 = 0$, $b_1 = -2$, $b_2 = -\frac{2}{2} = -1$, and $b_3 = -\frac{2}{3}$. So, you have

$$g(x) = \frac{2\pi}{2} - 2 \sin x - \sin 2x - \frac{2}{3} \sin 3x$$

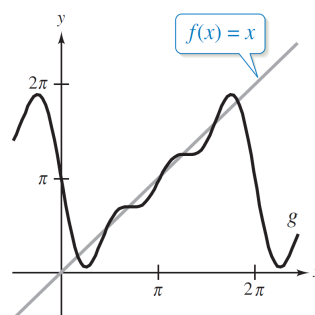
$$= \pi - 2 \sin x - \sin 2x - \frac{2}{3} \sin 3x.$$

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Example 7 – Solution (2 of 2)

The figure shown below compares the graphs of f and g .



Third-Order Fourier Approximation

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